

Lab assignment - 4

Q. 1: To implement a Naïve Bayes classifier for categorical input features and evaluate its performance on a real dataset.

Dataset: Use the Play Tennis dataset (weather → play/no play) or Mushroom dataset from [UCI repository](#).

Tasks:

- A. Load the dataset and display the first few records.
- B. Convert categorical values into numerical codes if needed (e.g., Sunny=0, Overcast=1, Rain=2).
- C. Split the dataset into training and test sets.
- D. Train a Categorical Naïve Bayes classifier.
- E. Predict class labels for the test set.
- F. Evaluate performance using accuracy, confusion matrix, and classification report.
- G. Discuss the effect of smoothing (Laplace correction).

Q 2: To implement a Naïve Bayes classifier for numerical input features and evaluate its performance.

Dataset: Use the Iris dataset (numerical features: sepal length, width, etc.) or Wine dataset.

Tasks:

- A. Load the dataset.
- B. Explore and visualize data distribution.
- C. Split dataset into training and testing sets.
- D. Train a Gaussian Naïve Bayes classifier.
- E. Predict and evaluate using accuracy, confusion matrix, classification report.
- F. Compare results with different random splits.

Q 3: In a two-class, two dimensional classification task, the feature vectors are generated by two normal distributions having the covariance matrices:

$$\Sigma_1 = \begin{bmatrix} 1.1 & 0.3 \\ 0.3 & 1.9 \end{bmatrix} \text{ and } \Sigma_2 = \begin{bmatrix} 1.0 & 0.1 \end{bmatrix}$$

and the mean vector are $\mu_1=[0, 0]^T$ and $\mu_2=[3,3]^T$ respectively. Classify the vectors [1.0, 2.2] according to the Bayesian classifier.

Note:

Laplacian Smoothing (Naïve Bayes)

In Naïve Bayes classification, we calculate probabilities such as:

$$P(\text{feature} \mid \text{class}) = \text{count}(\text{feature}, \text{class}) / \text{count}(\text{class})$$

But sometimes a feature never appears in the training data for a given class, making the probability zero. Since Naïve Bayes multiplies probabilities, a single zero makes the entire prediction probability zero, even if other features strongly suggest that class.

Solution: Laplace Smoothing

To fix this, we add 1 to all counts so that no probability becomes zero.

Formula with smoothing (for categorical features):

$$P(\text{feature} \mid \text{class}) = (\text{count}(\text{feature}, \text{class}) + \alpha) / (\text{count}(\text{class}) + \alpha * k)$$

Where:

- α = smoothing parameter (usually 1, called Laplace smoothing; if fractional, called Lid stone smoothing)
- k = number of possible categories for that feature

Example

Suppose we have a Play Tennis dataset and want $P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes})$.

Counts:

- Count (Sunny, Yes) = 0 (never observed in training)
- Count (Yes) = 9 (total 'Yes' examples)
- Possible outlook values = 3 (Sunny, Rain, Overcast)

Without smoothing: $P(\text{Sunny} \mid \text{Yes}) = 0 / 9 = 0$

With Laplace smoothing ($\alpha = 1$): $P(\text{Sunny} \mid \text{Yes}) = (0+1) / (9 + 3) = 1/12 \neq 0$